Framework for Blockchain-Based Community-Assisted Privacy- Preserving Reporting Service for Local Governance

An undergraduate project proposal report submitted to the

Department of Electrical and Information Engineering
Faculty of Engineering
University of Ruhuna
Sri Lanka

in partial ful llment of the requirements for the

Degree of the Bachelor of the Science of Engineering Honours

by

K.A.L.G. ATHTHANAYAKA - EG/2021/4419
J.S. KARUNARATHNA - EG/2021/4599
A.D.H. KARUNATHILAKE - EG/2021/4604
P.A.M.J.A. PATHIRAJA - EG/2021/4703

.....
Dr. Subodha Gunawardena
(Supervisor)

Abstract

Local governments in developing countries often struggle to address public issues e ciently. These struggles are mainly due to limited transparency, centralized data management, and insu cient citizen participation. E-governance and crowd-sourced reporting systems are designed to enhance public engagement. But their e ectiveness is limited by issues related to data integrity, user anonymity, content reliability, and trustworthiness.As a solution to the above challenges, we propose a blockchain-based, community-assisted, privacy-preserving reporting framework for local governance. The main objective of the project is to design and prototype a permissioned blockchain-based reporting system. This system will enable citizens to anonymously submit public issue reports and participate in opinion polling. Also, our system will ensure data immutability and transparency. To manage reporting work ows, issue tracking, and community voting in a secure and tamper-resistant manner, the blockchain will use Proof-of-Authority (PoA) mechanism and smart contracts. User identity is decoupled from the report content using a simulated identity and an access control mechanism to preserve the privacy of the users. Large data objects, such as images, are stored o -chain with cryptographic hashes anchored on the blockchain. Additionally, an Arti cial Intelligence (AI)-assisted content moderation mechanism will be used to minimize the impact of malicious, low-quality reporting content. To demonstrate the feasibility of the proposed framework, A web-based Decentralized App (DApp) is developed as a proof of concept within a controlled experimental environment. Through this project, we aim to deliver a practical and scalable prototype that enhances citizen participation, accountability, and trust in local governance systems.

Acknowledgments

Acknowledgments goes here.

Contents

Abstract i

Acknowledgements ii

Contents v

List of Figures vi

List of Tables vii

Acronyms viii

1 Introduction 1
1.1 Problem Background 1
1.2 Problem Statement 2
1.3 Objectives and Scope 3
1.3.1 Aim 3
1.3.2 Objectives 3
1.3.3 Scope 3
1.4 Literature Review 4
1.4.1 Digital Citizen Reporting and e-Governance Systems 4
1.4.2 Blockchain for Local Governance Applications 5
1.4.3 Blockchain-Based Citizen Reporting and Complaint Management Systems 5
1.4.4 Permissioned Blockchains and PoA Consensus Mechanisms . 7
1.4.5 Smart Contracts for Public Issue Management and Opinion Polling 8
1.4.6 User Interfaces for Blockchain-Based e-Governance Applications 9
1.4.7 AI-Based Content Moderation in Civic Reporting 11
1.4.8 Need for Automated Content Moderation 11
1.4.9 Architectural Approaches for AI Blockchain Integration . . . 11

1.4.10 AI Oracles and Decentralized Oracle Networks 12
1.5 Report Outline 13

2 Background 14
2.1 Blockchain Technology 14
2.2 Permissioned Blockchain and PoA 14
2.3 Smart Contracts 14
2.4 Decentralized Applications (DApps) 14
2.5 InterPlanetary File System (IPFS) 14
2.6 Zero-Knowledge and Identity Concepts 15
2.7 AI-based Content Moderation 15
2.8 Docker and Containerized Deployment 15

3 Methodology and Implementation 16
3.1 Methodology 16
3.1.1 Research Methodology 16
3.1.2 System Requirements Analysis 16
3.1.3 Proposed System Architecture 16
3.1.4 Proposed Work ow 16
3.1.5 Project Plan and Timeline 16
3.2 Implementation 17
3.2.1 Permissioned Blockchain Implementation 17
3.2.2 Smart Contract Implementation 17
3.2.3 Identity and Access Control Implementation 17
3.2.4 IPFS O -chain Storage Integration 17
3.2.5 AI-based Moderation System Implementation 17
3.2.6 Web-based DApp Development 17
3.2.7 Deployment Environment 17

4 Testing / Experimentation / Validation / Veri cation 18
4.1 Experimental Setup 18
4.2 Testing Methodology 18
4.3 Blockchain Performance Testing 18
4.4 Smart Contract Validation 18
4.5 AI Moderation Testing 19
4.6 IPFS Storage Testing 19
4.7 End-to-End Integration Testing 19
4.8 Results and Observations 19
4.9 Comparison with Existing Systems 19

5 Discussion 20
5.1 Novel Contributions of the Project 20
5.2 Overall System Validity 20
5.3 Advantages of Proposed Architecture 20
5.4 Limitations of Current Prototype 20
5.5 Real-world Applicability 21

6 Conclusion and Future Work 22
6.1 Conclusion 22
6.2 Future Work 22

A Smart Contract Code 23

B Oracle API Endpoints 24

C Docker Compose Files 25

D System Diagrams 26

E Testing Screenshots 27

F GitHub Repository Links 28

Bibliography 29

List of Figures

1.1 System Architecture of the Police Complaint Management System.[?] 6

List of Tables

Acronyms

AC - Alternating Current
DC - Direct Current
LV - Low Voltage
MV - Medium Voltage

Chapter 1

Introduction

Local governance is fundamental to the upkeep of public infrastructure and the overall well-being of the community. To address public problems such as damaged roads, issues in waste management, street light failures, e ective communication between citizens and local authorities is critical. Recently, the integration of e-governance systems and crowdsourcing has improve the citizen participation and streamline the reporting and resolution of public issues.

However, most of the current systems rely on centralized architectures. This poses a challenge in dealing with data integrity, transparency, and user privacy. Citizens are reluctant to provide complaints about sensitive topics or criticism due to potential retaliation or the possibility of their personal info being abused. At the same time, the local government is faced with the challenge of verifying the authenticity of anonymous complaints raised by citizens.

Some of the key features of Blockchain technology are immutability, transparency, and decentralization. Those features can be used to address several limitations of conventional reporting systems. It is possible to design reporting platforms that protect user anonymity while ensuring the integrity and accountability of reported information by combining blockchain-based architectures and privacy-preserving mechanisms. This project examines the application of blockchain technology in developing a secure, community-driven reporting framework tailored to local governance contexts.

1.1 Problem Background

In the context of local governance, crowdsourced reporting systems and e-governance platforms are widely adopted. By using those systems, citizens can report issues, provide feedback and participate in decision making. It improves the inclusiveness and the accountability of every citizen. In developing countries like Sri Lanka, these systems have the potential to signi cantly improve the governance e ciency.

Timely identi cation and the resolution of public conccerns are the key to governance e ciency.

However, despite their advantages, the conventional e-governance and reporting systems are still highly centralized, making them prone to various threats such as data tampering, illegal access, and single points of failure. In addition, the ability to track behavior down to individual users could serve as a discouraging factor, particularly in reporting delicate matters, in a system that lacks anonymity. Blockchain-based systems have appeared as a feasible solution for managing shared data in a tamper-proof and transparent manner. The ability of permissioned blockchains to enable controlled participation while maintaining accountability through trusted validators makes them a desirable choice for governance applications. By using smart contracts, reporting work ows and opinion polling can be automated rather than relying on centralized intermediaries.

While blockchain technology o ers data integrity and transparency, it still poses challenges related to user privacy, content moderation, and trustworthiness among anonymous participants. These challenges remain open research and implementation problems. Combining blockchain with privacy-preserving identity mechanisms and AI-assisted moderation can help address these limitations and supports the need for a comprehensive framework for secure and reliable community-assisted reporting.

1.2 Problem Statement

The existing frameworks of e-governance and crowdsourced reporting systems for local governance do not provide an integrated mechanism that can guarantee data integrity, user anonymity, and trustworthiness of citizen submitted reports. Centralized architectures are prone to unauthorized access, data tampering, and loss of transparency. At the same time, identity dependent systems discourage citizen engagement due to privacy concerns and fear of retaliation.

Although blockchain has been applied to the context of local government context, many blockchain-based reporting systems still fail to balance anonymity with mechanisms for assessing the reliability and quality of submitted reports. Furthermore, the lack of content moderation and trust evaluation increases the risk of misinformation, and malicious or low quality submissions.

Therefore, there is a clear need for a privacy-preserving, blockchain-based reporting framework that enables anonymous citizen participation while ensuring data immutability, transparent work ows, and reliable evaluation of reported information. This project aims to address this gap by designing and prototyping a permissioned blockchain-based community-assisted reporting system that incorporates smart contracts, o -chain data storage, and AI-assisted content moderation

mechanism suitable for local governance environments.

1.3 Objectives and Scope

1.3.1 Aim

The aim of this project is to design and prototype a blockchain-based, privacy-preserving, community-assisted reporting system for local governance that empowers citizens to report public issues and provide their opinions in a transparent, accountable, and secure manner.

1.3.2 Objectives

1. Design and implement a permissioned reporting blockchain using a PoA consensus mechanism to ensure the integrity and immutability of citizen-submitted reports.

2. Design and deploy smart contracts to automate public issue reporting workflows, including ticket creation, issue resolution and closure, community voting as feedback on resolved issues, and separate opinion polling mechanisms for public consultation.

3. Develop user-friendly interfaces that enable citizens and authorized government officials to interact with the blockchain for reporting issues and tracking their resolution status.

4. Design and integrate an AI-based content moderation system to detect and filter inappropriate textual and visual content in user-submitted reports.

1.3.3 Scope

1. A prototype of the proposed blockchain will be implemented with a minimum of three full nodes. The PoA consensus mechanism will be used to enable efficient and controlled transaction validation within a permissioned network, while excluding deployment of public, permissionless blockchains or globally distributed ledger infrastructures.

2. User authentication and identity decoupling are managed through a simulated identity service, and the design and implementation of a fully decentralized Self-Sovereign Identity (SSI) blockchain are explicitly excluded from the scope of this project.

Susceptibility to data tempering, lack of transparency are the main problems that impact heavily on public trust. In addition to that, mandatory identity disclosure is discouraging public participation on this kind of reporting systems. Therefore these problems motivate the ndings of new decentralized, transparent and immutable architectures that still preserve user privacy with accountability.

1.4.2 Blockchain for Local Governance Applications

Blockchain is a technology that has been popular for its ability to have an immutable, transparent and decentralized ledger for public data. Hence, its a transformative tool for e-governance to address the problems that existing implementations had. The Blockchain allow us to verify the immutability of the records, so its suitable for governance applications where the trust and the accountability is crucial. [?], [?].

Blockchain based land registry system implemented in Cook Country, Illinois is a one real world example of the applicability of blockchain in local governance. They aim to reduce the property fraud by consolidating historical ownership records into a veri able block in the blockchain. Similar to that Delaware has taken initiative to explore the use of smart contracts that's enable real time asset tracking and automate business llings. This improves the regulatory compliance and oversights [?].

In addition to administrative e ciency, blockchain has been utilized for citizen centric services, such as secure digital identity management. Estonia has taken a initiative to integrate blockchain across healthcare, taxation and even electronic voting systems. So this can provide a mathematically veri able audit trail for government data. MyPass project in Austin and the World Food Program's building blocks initiative also use Blockchain technology under the hood to manage encrypted identity and transaction records [?].

Despite the advancements in the blockchain base applications, existing literature highlights the challenges of adoption in governance. Scalability constraints, privacy concerns related to identity data, performance overhead and also the need of the legislative and institutional reforms are some examples. These factors needs to be addressed by carefully selecting blockchain architectures and consensus mechanisms before implementing systems for frequent citizen interactions.

1.4.3 Blockchain-Based Citizen Reporting and Complaint Management Systems

The traditional centralized platforms have data integrity issues, transparency issues and trust issues when it comes to citizen reporting. Past works also indicates that centralized complaint registries often have single point of failures , delayed

responses and public mistrust. Therefore blockchain integrated reporting services can be used to mitigate those issues.

A blockchain based police complaint management system has proposed by Pratheeba [?] contains a modular architecture. It's include security, blockchain, web interface and a hybrid cloud storage. This system ensure immutability of complaint records by cryptographic hash linking. And also it utilize RSA based encryption to restrict the access only to authorized personnel. In Addition to that, Pratheeba used a Natural Language Processing (NLP) techniques to automatically categorize complaints in the system based on their priorities.

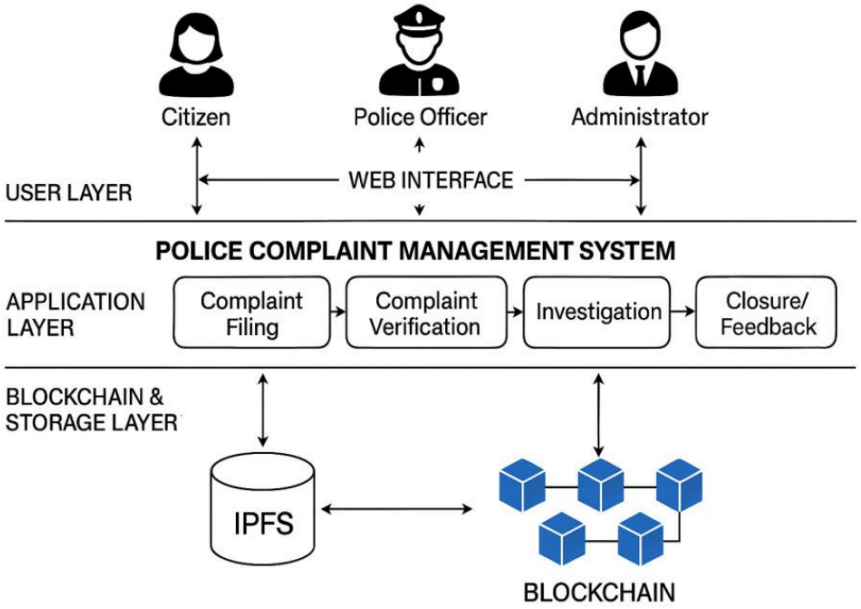

Figure 1.1: System Architecture of the Police Complaint Management System.[?]

A three tier architecture for decentralized complaint management framework has proposed by Manjula [?] is shown in the Figure 1.1. The three layers are web based presentation layer, a business logic layer and nally a blockchain data layer. This system has a role based access control method to involve multiple stakeholders. Citizens, Law enforcement authorities and judicial bodies are some of them. More importantly this system has a immutable storage for multimedia evidences. Intelligent complaint assignment based on workloads, automatic escalation mechanism are some of the key features of the framework. The experimental evaluations shows that a signi cant reductions in complaint processing time. The successful detection of data tampering also archived by blockchain veri cation.

In Addition to that PoA avoids the indeterministic transaction fees and support controlled validation environments. This make them viable for handling sensitive governmental data under regulatory constraints[?],[?]. As a result, PoA emerges as a practical consensus mechanism for permissioned blockchain-based reporting systems in local governance contexts.

1.4.5 Smart Contracts for Public Issue Management and Opinion Polling

In governance systems, to improve the transparency, e ciency and citizen participation, smart contracts have been explored as the good mechanism. Smart contracts can automate the governance rules and incentives by immutable code. It can support models like civic intelligence governance (CIG). In CIG, citizen engagement is encouraged through on chain participation and token based incentives[?]. Literature shows that approaches like CIG can improve participation rates and knowledge sharing while reducing relances on centralized intermediaries. In the context of multi organizational governance smart contracts also been used within dual blockchain architectures. This method separates transaction data from contract logic. It balance the transparency and privacy in Government to Government and Business to Business processes[?].

Smart contracts also support business processes in governance by embedding policy rules directly into executable logic. Bid submission, evaluation and contract execution are some of the use cases in the public procurement and tendering processes that smart contract hold its value. It also reduce the oppertunities for corruption and increase the accountability[?], [?], [?]. Similar bene ts are observed in the blockchain based land registry systems. Smart contracts have enable immutable property records automated ownership transfers and enforcement of land use policies. This has reduced fraud and administrative ine ciencies[?], [?].

In the context of voting and opinion polling, smart contracts o er enhanced security, transparency, immutability and veri ability inherited from blockchain[?], [?]. There are advance mechanisms such as Quadratic Voting and also blockchain enabled participatory budgeting. Those mechanisms further extend citizen involvement in decision making. It have expressive preference representation and transparent allocation of public resources[?], [?]. In addition to these advantages, there exists some challenges related to scalability, privacy and system complexity when deploying voting and polling systems to large and diverse populations[?], [?].

1.4.6 User Interfaces for Blockchain-Based e-Gover nance Applications

User interfaces play a critical role in determining the accessibility and adoption of blockchain based e-governance systems. In the traditional applications the user literacy is less important but in this conventional blockchain applications the user literacy is highly considered because of the blockchain concepts that are di erent from the traditional apps and it does make the users hard to understand and adopt. The literature consistently identi es usability and user experience challenges as major barriers to the adoption of decentralized applications for the general public. Particularly where security requirements con ict with established usability principles [1], [2]. So the design of intuitive, inclusive, and responsive web interfaces is essential for enabling e ective interaction between citizens, government o cials, and the underlying blockchain infrastructure.

Usability Challenges in Decentralized Governance Systems

Decentralized applications are fundamentally di erent from traditional web systems due to its architectural characteristics. Blockchain based systems have asynchronous transactions, immutable records, and cryptographic identity management. They can create friction for users who are used to centralized platforms. Studies highlight a signi cant usability gap in Web3 applications because of the concepts like transaction nality, gas fees, and private key ownership which are di cult for non technical users to understand and manage [1], [2].

Identity Management and Cognitive Load

One of the most critical usability barriers is wallet based identity management. The reason is user identity is tied to cryptographic key pairs instead of institution managed accounts In decentralized systems. Research shows that the responsibility of safeguarding private keys and the unrecoverble key loss red ce the user participation. Specially in general public where inclusivity is essential [2], [3], [4].

The literature further shows that concepts such as self-sovereign identity and identity management remain poorly understood by most users. Studies report unsafe practices and misunderstanding of recovery mechanisms when users are presented with seed phrases or key-based authentication [5], [6]. These ndings motivate the need for interface-level abstractions that preserve blockchain security while reducing cognitive load for citizens and o cials.

Transaction Latency and User Feedback

Blockchain transactions are inherently asynchronous with con rmation times dependent on network conditions and consensus protocols. This latency can lead to confusion among the users because of the system status problems and users can determine those as system failures [2]. The literature recommends to use an optimistic user interface pattern, where user actions are acknowledged immediately while blockchain con rmation occurs in the background.

As advanced design solutions, literature shows progressive status disclosure, distinguishing between o -chain receipt of a report and on-chain veri cation which allows users to track the lifecycle of an issue without being exposed to unnecessary low level blockchain details [7], [8]. Mechanisms like that are particularly relevant for issue reporting systems that require transparency and trust.

Transaction Costs and Abstraction of Blockchain Complexity

A signi cant usability challenge coming from the blockchain is transaction cost that associated with public blockchains. And also the concepts of gas fees and uctuating gas costs are very unfamiliar in public services context. Most of the time users expect free access to the reporting platforms[1]. The literature strongly suggest that the use of gas fee abstraction techniques. Meta Transactions is one of the example. In Meta, backend relayer services submit transactions on behalf of the user. Its achieved by preserving cryptographic intent veri cation[2]. Therefore this approach enable citizen to interact with blockchain without technical or economical complexities.

Comparison with Centralized Reporting Platforms

FixMyStreet and SeeClickFIx are existing centralized platforms. Eventhough they are centralized, they provide a valuable benchmarks for crowdsource citizen reporting interfaces. Inituitive map based reporting, simple authentication mechanisms and clear feedback loops are the key characteristics that attributed to platform success[9], [10].

With addition to that SeeClickFix further uses a gami cation mechanism to encourage the users to participate in the system. In contrast, Even though the proposed application uses blockchain for reporting platform, it must retain the usability and the expectations of the users. The literature highlights the need of interface design such that gurantees the data integrity and auditability using familiar ui elements rather than cryptographic representations[11], [8], [12].

Accessibility and Inclusivity Considerations

Inclusivity is a fundamental requirement for e-governance systems. According to [13], it can be seen that poorly designed blockchain interfaces may digitally divide users. Also it may favor crypto-literate users excluding vulnerable populations. To reduce this issue, literature suggests mobile rst design approaches. Furthermore, lightweight client architectures, e ecient bandwidth synchronization can used to support access in resource limited environments. [14].

1.4.7 AI-Based Content Moderation in Civic Reporting

Since blockchain is inheritably immutable its introduce a unique challenge for content moderation in a civic reporting system. In a centralized platforms, administrators can remove or edit inappropriate content. But in a decentralized blockchain based system data cannot be altered once con rmed [15]. While immutability strengthens transparency and accountability, it also raises the risk of permanently storing harmful, illegal, or misleading content. Therefore the literature focuses the need for pre moderation strategies to prevent unsuitable content from being recorded on chain or to limit its visibility after publication [7].

1.4.8 Need for Automated Content Moderation

Civic reporting platforms are susceptible to several forms of misuse that can compromise system reliability and public trust. Automated spam submissions, the submission of malicious or legally sensitive content, and deliberate dissemination of false or misleading reports are some of the common issues [16], [17], [18]. Manual moderation is impractical for above situations. Since the reporting system is large latency, human bias, and operational costs can occur. Furthermore it reintroduces the centralized control structure that blockchain system is aim to minimize.

Therefore the literature supports AI assisted moderation as a scalable and e cient approach for ltering content before its permanently recorded. It maintain the acceptable levels of decentralization [19], [20].

1.4.9 Architectural Approaches for AI Blockchain Integration

Due to the computational and storage limitations of blockchain platforms, AI-based moderation cannot be performed directly on chain. O chain processing combined with on chain veri cation identi ed by existing researches as most practical approach[20], [21].

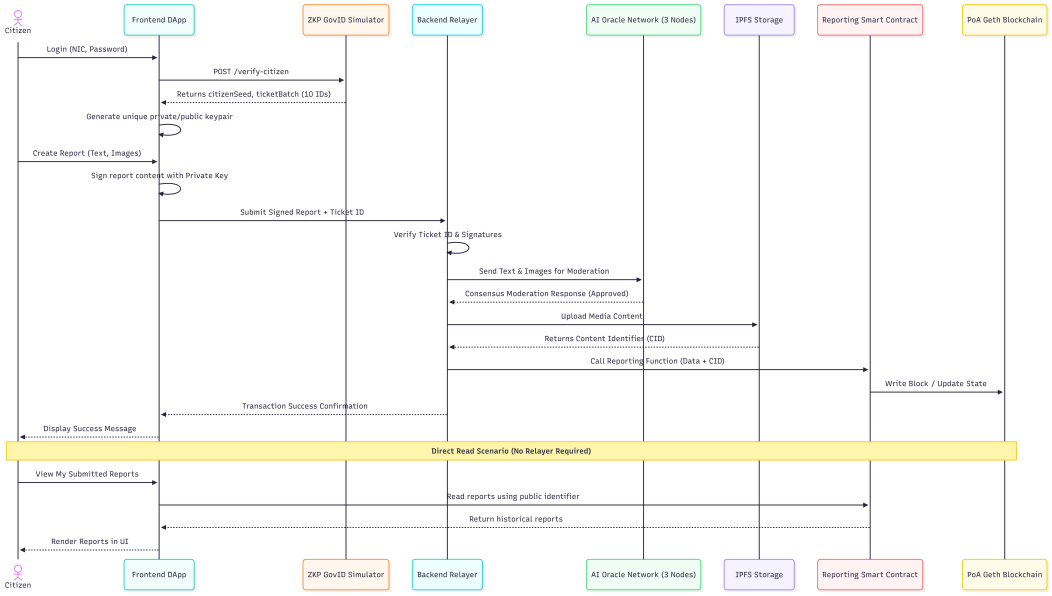

Figure 2.2: Sequence diagram for the report submission work ow.

2.5 Security and Privacy Considerations

Discuss:

1. pseudonymous identity,
2. digital signatures,
3. ticket validation,
4. hash integrity,
5. o -chain storage,
6. oracle voting.

[24] K. Zintus-art, B. Vass, and J. Ward, \Empirical evidence in AI oracle development," March 21 2025. Chainlink Blog. Accessed on January 9, 2026.

[25] CodeZeros, \AI-powered oracles: How generative AI is enhancing blockchain data feeds," April 10 2025. Accessed on January 9, 2026.

[26] M. Prasad, \Blockchain + AI in combating deepfake content circulation," Scienti c Journal of Arti cial Intelligence and Blockchain Technologies, April 2024. Accessed on January 9, 2026.

[27] A. Kumar, \Hybrid moderation models: Balancing AI and human oversight," October 27 2025. Infosys BPM. Accessed on January 9, 2026.

[28] B. Kae, \Research: AI agent sector overview," March 10 2025. The Block Research. Accessed on January 9, 2026.

[29] Chase Advisors, \Ai and content moderation," tech. rep., Chase Advisors, January 2024. Accessed on January 9, 2026.

[30] K. Mounika and N. R. Reddy, \An integrated machine learning framework for spammer and fake user detection in online social networks," Fringe Management, Economics and Politics, 2024. Accessed on January 9, 2026.